# ORRIN VALLEY STEEL WORKS

Orrin Valley · Certified Mill Test Report (synthetic)

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**Certificate No: MTR-OV-58415**
Heat No: OV-58415
Grade: 4140  Condition: Annealed  Product: Round bar, 2.000 in diameter
Specification: HLS-4140 Rev C
Customer: Brannock Metals Supply  Customer PO: BMS-31977

**CHEMICAL ANALYSIS (heat, wt %)**

| C | Mn | P | S | Si | Cr | Mo |
|---|---|---|---|---|---|---|
| 0.390 | 0.880 | 0.012 | 0.018 | 0.240 | 1.020 | 0.190 |

**MECHANICAL PROPERTIES**

Tensile strength: 98.7 ksi
Yield strength (0.2% offset): 66.1 ksi
Elongation in 2 in: 24 %
Hardness: 229 HB

**CERTIFICATION**

Material was melted and manufactured in the United States of America.
Material is free of mercury contamination; no mercury-bearing equipment was used in its manufacture.
We certify that this material was tested as reported above and that the results are correct.

**Signed: R. Okafor, Quality Assurance Manager**
Date: 2026-09-11